

FILTER DESIGN TEST CASE

Duplexer Module - WiFi 2.4 GHz for 4G LTE Carrier Aggregation
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Duplexer Module filter targeting WiFi 2.4 GHz for 4G LTE Carrier Aggregation. The filter is designed on AIN on SiC substrate with a target center frequency of 3592.7 MHz and bandwidth of 291.0 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 4G LTE Carrier Aggregation module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Duplexer Module
- Frequency Band: WiFi 2.4 GHz
- Application: 4G LTE Carrier Aggregation
- Substrate: AIN on SiC
- Package: LGA (Land Grid Array)
- Design Tool: PathWave EM Design
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the Duplexer Module design targeting WiFi 2.4 GHz. Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3592.7 MHz
Bandwidth (3 dB):	291.0 MHz
Insertion Loss Target:	≤ 2.1 dB
Return Loss Target:	≥ 9.0 dB
Out-of-Band Rejection:	≥ 45.0 dB
Isolation (Duplexer):	≥ 27.0 dB
Q Factor:	13907
Temperature Coefficient:	-8.0 ppm/°C
Die Size:	2.1 x 0.8 mm
Wafer Size:	8-inch
Number of Resonators:	6
Electrode Material:	Mo/Al

Passivation: SiO2

4. TEST PROCEDURE

1. Open PathWave EM Design and load the Duplexer Module template project.
2. Define the substrate stack: AlN on SiC with specified layer thicknesses.
3. Set design targets: center freq 3592.7 MHz, BW 291.0 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.1 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (LGA (Land Grid Array)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on SiC).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 10778 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.1 dB
- Return loss in passband shall be >= 9.0 dB
- Out-of-band rejection at +/- 436 MHz offset >= 45.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 3.6 MHz
- Power handling shall be >= +27 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 87%

6. TEST RESULTS

Measurement Date: 2024-06-04
Devices Tested: 95
Insertion Loss (meas): 1.84 dB
Return Loss (meas): 11.2 dB
Rejection (meas): 47.8 dB
Isolation (meas): 31.7 dB
Q Factor (meas): 13907
Group Delay Variation: 4.0 ns
Power Handling: +34 dBm
Die Yield: 91.7%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The Duplexer Module design demonstrated below-target performance for WiFi 2.4 GHz.
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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